Multimodal Large Language Models in Healthcare

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***Abstract*—Healthcare is a naturally multimodal activity: clinicians synthesize patient narratives, laboratory trajectories, imaging studies, waveforms, medications, genomics, prior encounters, and biomedical evidence before making decisions. Multimodal large language models (MLLMs) promise a more flexible form of medical artificial intelligence by connecting language models with visual, temporal, tabular, and retrieval components. Their appeal is not simply that they can classify an image or answer a question, but that they can communicate uncertainty, compare evidence across records, generate structured reports, and interact with clinicians through language. This survey provides a publication-oriented synthesis of MLLMs in healthcare, covering foundations, architectures, training objectives, datasets, benchmarks, specialty applications, deployment workflows, safety risks, regulation, and future research directions. We trace the field from contrastive medical vision–language representation learning through instruction-tuned biomedical assistants and recent generalist biomedical systems. We compare representative systems including CLIP, Flamingo, BLIP-2, InstructBLIP, LLaVA, GPT-4V, MedCLIP, BioViL, CheXzero, LLaVA-Med, Med-Flamingo, Med-PaLM M, Med-Gemini, and newer generalist biomedical vision–language models. We also examine evaluation gaps: commonly used leaderboards reward short-answer accuracy or lexical overlap, whereas clinical deployment requires factual consistency, calibration, subgroup robustness, evidence grounding, workflow impact, and prospective validation. The review argues that progress in healthcare MLLMs depends less on a single larger model than on reliable multimodal grounding, transparent data governance, specialty-aware evaluation, human-centered interfaces, privacy-preserving collaboration, and post-deployment monitoring under medical-device and health-AI governance frameworks.**

***Index Terms*—Multimodal large language models, biomedical artificial intelligence, medical vision–language models, clinical decision support, radiology report generation, medical visual question answering, trustworthy AI, healthcare regulation.**

I. Introduction

HEALTHCARE has never been a single-modality problem. A chest-pain workup may combine symptoms, electrocardiograms, troponin trends, chest radiographs, medication history, risk factors, and institutional protocols. A tumor board may combine pathology slides, radiology, genomics, operative notes, staging criteria, and patient preferences. In intensive care, a clinician may interpret waveforms, vitals, medication infusions, labs, ventilator parameters, progress notes, and imaging studies under time pressure. This evidence is not merely stored in different formats; it carries different uncertainty, temporal granularity, spatial scale, and clinical authority. Traditional medical AI systems have therefore been useful but narrow: a classifier predicts a label, a segmenter outlines an organ, and a risk model estimates a probability. The result is often a collection of useful but disconnected tools rather than an assistant capable of synthesizing clinical context.

MLLMs change the design center of healthcare AI. They pair a large language model (LLM) with encoders for images, video, waveforms, tables, or external knowledge, and they expose the resulting system through an instruction-following interface. General vision–language models (VLMs) such as CLIP, ALIGN, Flamingo, BLIP, BLIP-2, InstructBLIP, LLaVA, GPT-4V, Gemini, and GPT-4o established the modern vocabulary of contrastive alignment, interleaved multimodal prompting, visual instruction tuning, and general-purpose perception-language reasoning [1]–[10]. Medical systems such as ConVIRT, GLoRIA, BioViL, MedCLIP, CheXzero, LLaVA-Med, Med-Flamingo, Med-PaLM M, Med-Gemini, and generalist biomedical VLMs adapt those ideas to radiology, pathology, ophthalmology, dermatology, clinical notes, and multimodal biomedical benchmarks [11]–[20].

The promise is substantial. An MLLM can read an image and a question, draft a report, compare current and prior examinations, retrieve guideline passages, transform free text into structured summaries, and communicate results in a form clinicians can inspect. However, the risks are also amplified. An image classifier may be wrong; a fluent MLLM may be wrong in a way that sounds coherent, cites irrelevant evidence, and encourages automation bias. A report generator may omit a small pneumothorax while producing polished prose. A triage assistant may combine biased historical data with an overconfident recommendation. Clinical deployment therefore demands more than benchmark accuracy: it requires evidence grounding, calibration, failure-mode analysis, privacy controls, auditability, human oversight, and regulatory alignment [21]–[24].

A. Scope and Review Method

This survey focuses on MLLMs and closely related biomedical VLMs that integrate language with at least one non-text clinical or biomedical modality. The review includes four families of work: (i) general-purpose multimodal foundation models that define reusable architectural patterns; (ii) biomedical representation models trained with image–text or report supervision; (iii) instruction-tuned medical assistants for visual question answering (VQA), report generation, and multimodal reasoning; and (iv) evaluation, governance, and deployment literature needed for clinical translation. We emphasize radiology because it is the most mature pairing of medical images and language, but we also discuss pathology, ophthalmology, dermatology, critical care, electronic health records (EHRs), genomics, education, and patient communication.

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The literature base was assembled from foundational machine-learning papers, biomedical AI journals, medical imaging conferences, open benchmark reports, official regulatory and governance documents, and recent arXiv or technical reports where peer-reviewed alternatives are not yet available. We prioritize primary sources and high-impact surveys, and we distinguish peer-reviewed claims from emerging preprints where appropriate. The objective is not to rank systems by a single leaderboard. Instead, the survey asks how design choices interact with clinical data, evaluation, workflow, safety, and institutional deployment.

B. Contributions

This manuscript makes six contributions. First, it reorganizes the original draft into an IEEE-style survey with a full technical taxonomy of healthcare MLLMs. Second, it expands the architecture discussion to include modality encoders, alignment modules, fusion strategies, retrieval and tool use, instruction tuning, parameter-efficient adaptation, and safety alignment. Third, it provides professional comparative tables for representative models, datasets, benchmarks, objectives, metrics, clinical applications, and deployment risks. Fourth, it introduces equations for attention, contrastive alignment, multimodal fusion, instruction tuning, preference optimization, calibration, and factuality-aware evaluation. Fifth, it critically analyzes clinical validation gaps rather than treating benchmark performance as a proxy for readiness. Sixth, it supplies an Overleaf-ready IEEEtran project with a BibTeX bibliography containing more than one hundred references and openly licensed figure/source documentation.

C. Organization

Section II reviews foundations of LLMs, VLMs, and clinical multimodality. Section III presents an architectural taxonomy and mathematical formulation. Section IV analyzes training, adaptation, and alignment. Section V synthesizes major literature streams. Sections VII and VIII examine datasets, benchmarks, and metrics. Section IX surveys applications and case studies. Sections X and XI discuss limitations, ethics, privacy, and regulation. Section XII outlines future directions, and Section XIII concludes the paper.

II. Background and Foundations

A. From Transformers to Instruction-Following Models

The Transformer introduced self-attention as a scalable way to model token interactions without recurrent computation [25]. Given input embeddings $X \in \mathbb{R}^{n \times d}$, a single attention head computes

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{1}$$

where $Q = XW_Q$, $K = XW_K$, and $V = XW_V$. LLMs extend this mechanism through large-scale autoregressive or masked-language pretraining, followed by supervised instruction tuning and preference alignment [26]–[30]. In healthcare, language-only models are useful for question answering, summarization, coding support, and biomedical literature retrieval,

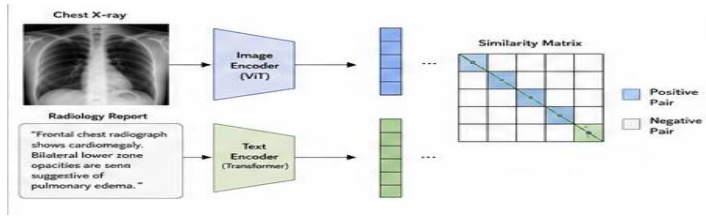

Fig. 1. Contrastive language–image pretraining schematic from the official CLIP repository, reused under the MIT license [1]. The same principle motivates many medical image–report pretraining methods, but clinical reports require special handling of negation, uncertainty, and multiple coexisting findings.

but they cannot directly interpret a radiograph, a histology slide, an ECG, or a retinal photograph.

Instruction-following is important because clinical usefulness depends on task framing. A pretrained model may encode knowledge but fail to produce safe answers in the style required by clinicians. Supervised fine-tuning teaches the model to map instructions to desired outputs, while reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) shape response behavior toward helpfulness, harmlessness, and truthfulness [28], [31]. In medicine, the target behavior is stricter: the model must know when to defer, express uncertainty, distinguish evidence from speculation, avoid patient-specific advice outside scope, and support clinician accountability.

B. Vision–Language Learning

Vision–language learning connects visual and textual representations. CLIP learns image and text encoders by maximizing agreement between matched image–caption pairs and minimizing agreement between mismatches [1]. For a batch of N image–text pairs, normalized embeddings v_i and t_i, and temperature τ, a common symmetric contrastive objective is

$$\mathbf{L}_{\mathrm{CL}} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{\exp(v_i^\top t_i/\tau)}{\sum_{j=1}^{N}\exp(v_i^\top t_j/\tau)} + \log\frac{\exp(t_i^\top v_i/\tau)}{\sum_{j=1}^{N}\exp(t_i^\top v_j/\tau)}\right]. \tag{2}$$

This objective enables zero-shot classification by comparing image embeddings with textual prompts, a property that strongly influenced biomedical contrastive models. However, medical image–text pairs are not simple web captions. A radiology report may describe absence, uncertainty, comparison with prior studies, tubes and lines, and clinically relevant negatives. Two different chest radiographs may share nearly identical findings, so treating all non-matching pairs as negatives can inject false-negative pressure. Biomedical VLP methods therefore adapt contrastive learning with domain-specific text encoders, local alignment, report semantics, or unpaired image–text strategies [11]–[14].

C. Healthcare Data Modalities

Clinical data are heterogeneous along at least five axes. First, data vary by sensor: CXR, CT, MRI, ultrasound, pathology, dermatology, ophthalmology, ECG, EEG, and bedside

waveforms. Second, they vary by structure: free-text notes, structured codes, numeric labs, medication administrations, images, and time series. Third, they vary by temporal scale: milliseconds for ECG waveforms, hours for vitals, days for inpatient notes, months for follow-up imaging, and years for chronic disease. Fourth, they vary by uncertainty: a laboratory value is numerically precise but clinically contextual, while a radiology impression may encode uncertainty and differential diagnosis. Fifth, they vary by governance: EHR data are protected health information, biomedical literature may be public, and imaging datasets may have reuse restrictions.

This heterogeneity is why simple concatenation is rarely sufficient. A clinically useful MLLM must maintain modality-specific structure while supporting cross-modal reasoning. For example, a CXR assistant should distinguish the image observation "small left pleural effusion" from the chart context "history of heart failure" and from a retrieved guideline statement. A pathology assistant must preserve cellular morphology while connecting it to molecular tests and report language. A generalist biomedical system must decide when a modality is relevant and when an absent modality makes a conclusion unsafe.

D. Generalist Biomedical AI

Generalist biomedical AI aims to build systems that can flexibly handle multiple modalities and tasks with shared parameters [18], [32]. Med-PaLM M introduced MultiMed-Bench, a benchmark spanning question answering, VQA, image classification, report generation, summarization, and genomics, and demonstrated that a unified generative system can compete with specialized baselines on diverse tasks [18]. Med-Gemini extended this direction through long-context reasoning, uncertainty-guided search, and specialized medical variants of Gemini [19]. A 2024 generalist biomedical VLM in *Nature Medicine* further underscored the shift from single-task medical models toward shared biomedical visual-language systems [20]. These systems are important, but they also sharpen a central tension: generality improves usability and transfer, while medicine often demands specialty-specific precision, calibration, and accountability.

III. Technical Architectures

A. Canonical Healthcare MLLM Pipeline

Most healthcare MLLMs can be described as a pipeline of modality encoders, alignment modules, a language backbone, and output controllers. Let $\mathcal{M} = \{m_1, \ldots, m_k\}$ denote modalities such as images, text, waveforms, and structured features. Each modality has an encoder E_m producing tokens or embeddings $Z_m = E_m(x_m)$. A projector, resampler, query transformer, cross-attention block, or adapter maps Z_m into a language-compatible token space:

$$\tilde{Z}_m = P_m(Z_m; \vartheta_m). \tag{3}$$

The LLM then conditions on textual instructions T, multimodal tokens $\tilde{Z}_{1:k}$, and optional retrieved evidence R:

$$p_\vartheta(y|T, x_{1:k}, R) = \prod_{t=1}^{|y|} p_\vartheta(y_t|y_{<t}, T, \tilde{Z}_{1:k}, R). \tag{4}$$

In practice, architectures differ in what is frozen, what is trainable, how many visual tokens are passed to the LLM, whether cross-attention layers are inserted, and whether external retrieval or tools are allowed.

B. Encoder Choices

Visual encoders include convolutional networks, Vision Transformers (ViTs), hybrid hierarchical transformers, and specialty encoders for whole-slide images (WSIs) or volumetric scans [33]. Chest radiographs can often be represented as 2-D images, but CT and MRI are inherently volumetric and may require slice sampling, 3-D encoders, or report-guided region selection. Pathology WSIs can contain billions of pixels, so models typically use patch extraction, multiple-instance learning, or hierarchical aggregation. Ophthalmology and dermatology require high-resolution local morphology and color fidelity. Signal encoders for ECG or bedside monitors require temporal filters, transformers, or state-space models that preserve rhythm and time intervals.

Text encoders vary from general biomedical language models such as BioBERT, PubMedBERT, ClinicalBERT, and BioGPT to the decoder backbones of large instruction-tuned LLMs [34]–[37]. In medical VLP, text encoder quality is not secondary. BioViL showed that better radiology-language modeling improves downstream visual grounding and segmentation even when the image objective is global [13]. This is intuitive: a report is not a generic caption but a structured clinical artifact with negation, uncertainty, comparison, and domain abbreviations.

C. Alignment Modules

Alignment modules bridge encoders and LLMs. The simplest approach uses a linear projector that maps visual features into the LLM embedding dimension, as in early LLaVA-style systems [7], [16]. BLIP-2 uses a Querying Transformer (Q-Former) to extract a fixed number of visual tokens from a frozen image encoder and interface with a frozen language model [5]. Flamingo inserts gated cross-attention layers into a frozen language model and uses a perceiver resampler to handle variable-length visual inputs [3]. Medical systems often favor lightweight adapters because institutional compute is limited and data are private. However, lightweight alignment can become a bottleneck when fine-grained spatial evidence is essential, as in small nodules, line placement, or pathology grading.

D. Fusion Strategies

Fusion is not a cosmetic design choice; it determines where modalities interact and how errors propagate. Early fusion combines modalities before deep reasoning, enabling dense interaction but requiring careful normalization and high memory. Late fusion processes modalities separately and combines decisions or embeddings near output, improving modularity and interpretability but limiting cross-modal dependencies. Hybrid fusion preserves modality-specific encoders and introduces

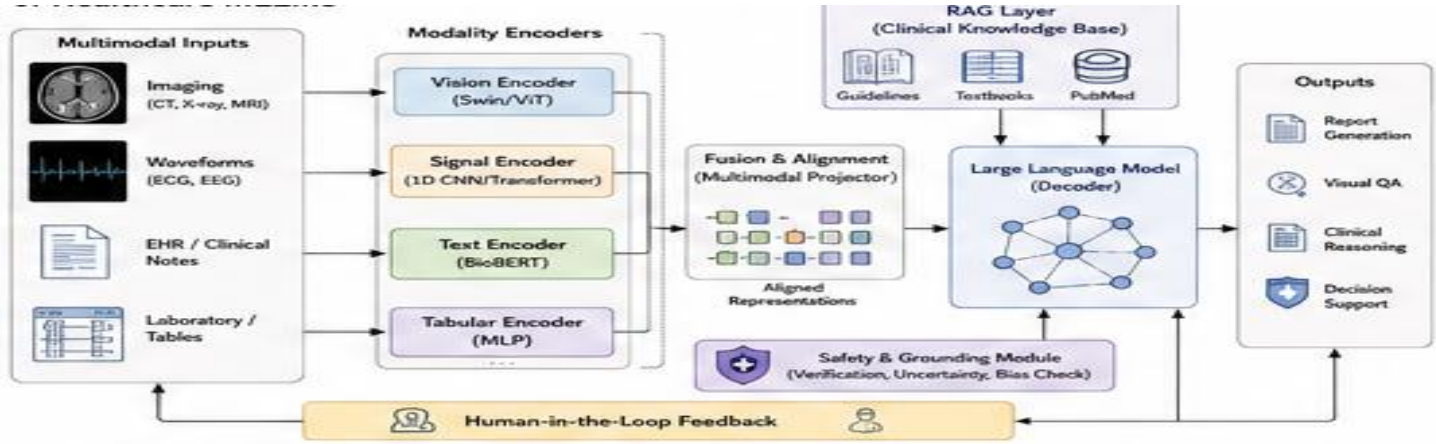

Fig. 2. Author-created architecture synthesis for healthcare MLLMs. The diagram separates modality-specific encoding, cross-modal alignment, language generation, retrieval/tool augmentation, safety controls, and human oversight. This separation is useful because clinical failure can originate in any layer, not only the LLM backbone.

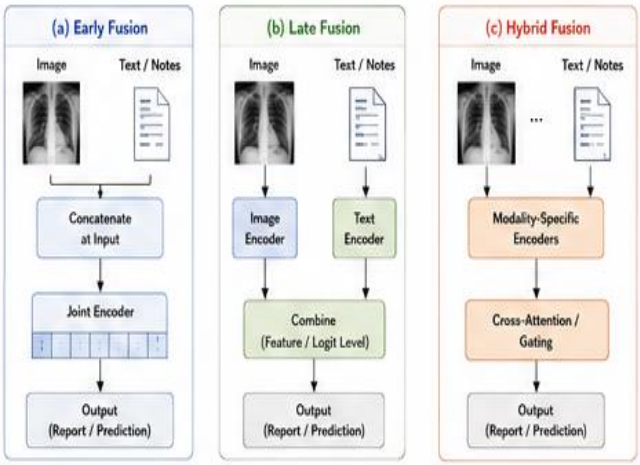

Fig. 3. Author-created comparison of early, late, and hybrid fusion. Hybrid fusion is common in healthcare MLLMs because it permits modality-specific preprocessing while still allowing language-guided cross-modal reasoning.

interaction through cross-attention, co-attention, adapters, or shared decoding. A generic hybrid formulation is

$$h = \phi\left(\sum_{m=1} \alpha_m A_m(E_m(x_m))\right), \quad \alpha_m = \mathrm{softmax}(g_m(E_m(x_m), \mathrm{pre}T)_n)_{\sigma t} \tag{5}$$

where A_m is an alignment operator, α_m is a context-dependent modality weight, and ϕ is the downstream reasoning module. In clinical settings, α_m should not be interpreted naively as explanation, but it can support auditing when combined with ablations and evidence localization.

E. Retrieval, Tools, and Structured Outputs

Pure parametric generation is poorly matched to medicine because clinical knowledge changes, local protocols differ, and patient-specific evidence must be traceable. Retrieval-augmented generation (RAG) conditions the model on external documents, similar cases, guidelines, prior reports, or biomedical literature [38], [39]. Tool-augmented systems may call calculators, rule engines, DICOM viewers, terminology services, or drug interaction databases. The challenge is evidence selection: a model must retrieve sources that are relevant, current, and appropriate for the patient context. In healthcare, retrieval should be coupled with provenance, versioning, and source-type labeling so clinicians can distinguish patient data, institutional policy, and public literature.

Structured outputs matter for interoperability. A report-generation assistant should not only produce prose; it may also output entities, uncertainty labels, measurements, RadGraph-style relations, or FHIR-compatible resources. Structured generation reduces ambiguity and enables downstream auditing, but it can also hide uncertainty if schemas are too rigid. A practical system should therefore preserve both structured fields and natural-language rationale, each linked to supporting evidence.

IV. Training, Adaptation, and Alignment

A. Pretraining Corpora and Data Construction

Healthcare MLLMs depend on data pipelines as much as architectures. A CXR corpus may pair images with reports, but the report may include comparison to prior studies that not available to the model. A biomedical figure-caption corpus extracted from PubMed Central contains useful broad-domain scientific imagery but also includes composites, plots, diagrams, microscopy, and publication-specific language that differ from routine clinical workflows [16], [40]. A pathology corpus may include slide-level diagnoses but not cell-level labels. EHR data are longitudinal and clinically rich, but they are difficult to de-identify and harmonize across institutions.

Data construction therefore requires explicit choices about granularity, linkage, de-identification, and quality control. Pair-level filtering removes broken image–text pairs; section parsing separates findings from impressions; labelers such

TABLE I
REPRESENTATIVE ARCHITECTURAL FAMILIES FOR HEALTHCARE MLLMS

Family	Core mechanism	Representative systems	Clinical strengths	Main limitations
Contrastive VLP	Dual encoders align image and text embeddings with contrastive loss	CLIP, ConVIRT, GLoRIA, BioViL, MedCLIP, CheXzero	Efficient retrieval, zero-shot labels, reusable encoders, label-efficient classification	Weak free-form reasoning, sensitivity to prompts, limited local grounding unless explicitly trained
Generative VLP	Visual tokens condition a language decoder for captions, reports, or answers	BLIP, BLIP-2, report-generation transformers	Natural-language outputs, report drafting, summarization	Fluency can mask factual errors; evaluation by lexical overlap is insufficient
Interleaved few-shot VLM	Frozen or lightly tuned LLM attends to interleaved images and text	Flamingo, Med-Flamingo	Few-shot adaptation, flexible demonstrations, multi-image context	High compute cost, limited transparency, possible domain mismatch
Visual instruction assistants	Projected visual tokens are paired with instruction-tuned LLMs	LLaVA, LLaVA-Med, medical LLaVA variants	Conversational VQA, flexible prompts, low-cost adaptation	Instruction data quality dominates behavior; may hallucinate or over-explain
Generalist biomedical systems	Unified sequence or multimodal architecture spans many biomedical tasks	Med-PaLM M, Med-Gemini, generalist biomedical VLMs	Broad task coverage, long-context reasoning, cross-domain transfer	Often closed or large-scale; clinical validation remains limited; evaluation is heterogeneous
Retrieval/tool-augmented clinical MLLMs	MLLM calls evidence stores, prior records, guidelines, calculators, or structured tools	Emerging clinical copilots, RAG medical assistants	Traceability, current knowledge, workflow-specific evidence	Retrieval errors, source conflicts, privacy constraints, harder regulatory boundary

as CheXpert, NegBio, and CheXbert extract disease labels from reports; graph tools such as RadGraph structure entities and relations [41]–[44]. For instruction tuning, some systems transform captions or reports into question–answer pairs using teacher LLMs, as in LLaVA-Med and PMC-VQA [16], [45]. This expands supervision but introduces teacher-model artifacts and requires clinical review when used for high-stakes tasks.

B. Pretraining Objectives

Objectives can be grouped into contrastive, generative, matching, grounding, and multitask losses. Contrastive loss aligns global embeddings as in (2). Image–text matching adds a binary objective to classify whether a pair corresponds. Captioning/report generation uses autoregressive likelihood:

$$L_{gen} = -\sum_{t=1}^{T} \log p_\vartheta(y_t | y_{<t}, x_{image}, T). \tag{6}$$

Grounding losses align text spans with regions, masks, or boxes. For region b and phrase s, a simplified grounding objective can combine classification and localization:

$$L_{ground} = L_{CE}(\hat{s}, s) + \lambda(1 - IoU(\hat{b}, b)). \tag{7}$$

Multitask biomedical systems combine objectives for VQA, classification, report generation, summarization, and genomics [18]. The design difficulty is balancing tasks with very different label noise and clinical stakes. A model should not improve a generic VQA score by degrading factual report generation.

C. Instruction Tuning

Instruction tuning converts a pretrained multimodal model into an assistant. A supervised instruction dataset $D = \{(I_i, x_i, y_i)\}_{i=1}^{N}$ contains instructions I_i, modality inputs x_i, and desired responses y_i. The standard loss is

$$L_{SFT} = -\sum_{i=1}^{N} \sum_{t=1}^{|y_i|} \log p_\vartheta(y_{i,t} | y_{i,<t}, I_i, x_i). \tag{8}$$

Medical instruction data should include multiple response styles: concise answers for VQA, structured reports for radiology, uncertainty-aware summaries for decision support, refusal and escalation for unsafe requests, and evidence-grounded rationales for audit. It should also include negative examples. A medical MLLM should learn not to invent missing patient history, not to infer protected attributes unless clinically justified, not to provide definitive treatment instructions to patients without context, and not to conceal uncertainty behind polished language.

D. Preference Alignment and Safety

Preference alignment can compare a preferred response y^+ and a dispreferred response y^- for the same prompt. DPO optimizes a policy relative to a reference model without explicitly training a reward model [31]:

$$L_{DPO} = -E \left[\log \sigma \left(\beta \log \frac{\pi_\vartheta(y^+|x)}{\pi_{ref}(y^+|x)} - \beta \log \frac{\pi_\vartheta(y^-|x)}{\pi_{ref}(y^-|x)} \right) \right]. \tag{9}$$

In healthcare, preference data should not merely ask which answer is more pleasant. It should distinguish factual correctness, clinical completeness, appropriate uncertainty, harm avoidance, evidence citation, and workflow fit. Radiologists may prefer a terse answer for a normal CXR but a structured differential for a complex chest CT. Patients may need plain-language explanations that avoid false reassurance. Preference alignment is therefore user-role dependent.

TABLE II
TRAINING OBJECTIVES AND THEIR HEALTHCARE IMPLICATIONS

Objective	Typical supervision	Benefit	Healthcare-specific risk
Contrastive alignment	Image–report or image–caption pairs	Label-efficient representations, retrieval, zero-shot classification	False negatives when different patients share findings; prompt sensitivity; weak temporal reasoning
Captioning/report generation	Image and target report text	Natural-language drafting and summarization	Fluent hallucination, omitted critical findings, lexical metrics misaligned with harm
Image–text matching	Matched and mismatched pairs	Improves pair discrimination and retrieval	Easy negatives may inflate performance; mismatches may be clinically similar
Grounding/localization	Regions, masks, boxes, phrase links	Supports visual evidence and auditing	Sparse annotations; heatmaps can be misleading without validation
Instruction tuning	Prompts, images, answers, reports, rationales	Conversational usefulness and task control	Synthetic instruction artifacts; unsafe compliance; role confusion
Preference alignment	Expert or user comparisons	Better response style, uncertainty, and safety behavior	Preferences vary by specialty and user role; costly expert labeling
Retrieval/tool tuning	Queries, evidence passages, tool traces	Traceability and current knowledge	Retrieval failure, stale guidelines, conflicting sources, audit complexity

E. Parameter-Efficient and Privacy-Preserving Adaptation

Hospitals often cannot fully fine-tune billion-parameter models on local data. Parameter-efficient fine-tuning (PEFT), especially LoRA, updates low-rank matrices inside frozen layers [46]. For a weight matrix W, LoRA represents the update as

$$W' = W + \Delta W, \quad \Delta W = BA, \quad \mathrm{rank}(BA) \ll \min(d_{\mathrm{in}}, d_{\mathrm{out}}). \tag{10}$$

PEFT makes local adaptation practical but does not solve governance. A small adapter can memorize local phrases, amplify institutional bias, or overfit to scanner-specific artifacts. Privacy-preserving strategies such as federated learning, split learning, differential privacy, and secure aggregation are promising, but they introduce statistical heterogeneity and operational complexity [47]–[49]. For MLLMs, privacy risk includes not only training data leakage but also prompt leakage, retrieved-document exposure, and generated summaries that combine identifiable facts.

V. LITERATURE REVIEW AND MODEL LANDSCAPE

A. General Multimodal Foundation Models

General VLMs supply reusable design patterns for healthcare. CLIP and ALIGN showed that large image–text corpora can train transferable dual encoders [1], [2]. BLIP and BLIP-2 introduced bootstrapped caption filtering and efficient bridging between frozen image encoders and frozen LLMs [4], [5]. Flamingo demonstrated few-shot learning over interleaved visual and textual context through gated cross-attention [3]. InstructBLIP and LLaVA showed that instruction tuning can turn a VLM into a conversational assistant [6], [7]. GPT-4V, Gemini, and GPT-4o then made multimodal interaction visible to a broad clinical and research audience [8]–[10].

These models are not medically reliable out of the box. Their training corpora underrepresent protected clinical data, their outputs may reflect general web priors, and their safety policies are not equivalent to clinical validation. Nevertheless, they establish mechanisms that medical systems reuse: contrastive alignment, visual token resampling, cross-attention, instruction-following, long-context prompting, and multimodal chain-of-thought style reasoning. The best medical systems often combine these general patterns with domain-specific data, evaluation, and guardrails.

B. Biomedical Vision–Language Representation Learning

Before conversational MLLMs, biomedical VLP focused on representation learning. ConVIRT learned medical visual representations from paired images and reports, improving label efficiency for CXR tasks [11]. GLoRIA aligned global and local image-text features, improving localization-sensitive recognition [12]. BioViL improved biomedical text semantics and introduced phrase grounding resources [13]. BioViL-T exploited temporal structure in current and prior radiology studies [50]. MedCLIP addressed the scarcity of paired image–text data by learning from unpaired medical images and texts with semantic matching [14]. CheXzero showed that CXR pathologies can be detected from unannotated image–report pairs through zero-shot prompt-based classification [15]. BiomedCLIP scaled scientific image–text pretraining using PubMed Central-derived pairs [51].

These systems reveal a recurring lesson: medical language is not an accessory to medical images; it is a supervision source, a source of bias, and an evaluation target. Reports encode observations, uncertainty, and institutional practice. If a model learns from reports, it may learn both useful radiological semantics and local reporting habits. The representation-learning phase therefore creates downstream behavior that may not be visible from a single benchmark score.

C. Medical Visual Instruction Models

LLaVA-Med adapts the LLaVA recipe to biomedicine by using biomedical figure-caption data and GPT-4-generated instructions, then fine-tuning in a curriculum that first aligns biomedical vocabulary and then learns conversational semantics [16]. Med-Flamingo adapts OpenFlamingo to medical image–text data and emphasizes few-shot VQA under scarce supervision [17]. PMC-VQA and MedVInT reframe medical VQA as generative instruction following and construct a large

TABLE III
REPRESENTATIVE MODELS AND SYSTEMS RELEVANT TO HEALTHCARE MLLMS

Model/system	Year	Modality focus	Key contribution	Healthcare relevance
CLIP [1]	2021	Image–text	Web-scale contrastive alignment and zero-shot prompting	Foundation for prompt-based medical image classification and retrieval
Flamingo [3]	2022	Interleaved image/video–text	Few-shot VLM with gated cross-attention	Pattern for few-shot medical VQA and multi-image context
BLIP-2 [5]	2023	Image–text–LLM	Q-Former bridges frozen visual and language models	Efficient medical adaptation with limited compute
LLaVA [7]	2023	Image–instruction	Visual instruction tuning using synthetic conversations	Template for LLaVA-Med and open medical assistants
ConVIRT [11]	2022	CXR–report	Contrastive medical visual representation learning	Improved label efficiency for CXR tasks
GLoRIA [12]	2021	CXR–report	Global-local image–text alignment	Better fine-grained disease localization
BioViL [13]	2022	Radiology image–text	Domain-specific text semantics and phrase grounding	Stronger radiology representations and grounding
MedCLIP [14]	2022	Medical image–text	Contrastive learning from unpaired medical data	Reduces dependence on paired image-report corpora
CheXzero [15]	2022	CXR–report	Zero-shot CXR pathology classification from unannotated pairs	Demonstrates clinical value of report supervision
LLaVA-Med [16]	2023	Biomedical image–text	Biomedical visual instruction tuning in one day	Conversational biomedical VQA and research-image interpretation
Med-Flamingo [17]	2023	Medical image–text	Few-shot medical VLM adapted from OpenFlamingo	Useful where labeled medical examples are scarce
Med-PaLM M [18]	2024	Multimodal biomedical	MultiMedBench and generalist biomedical AI	Unifies QA, imaging, report generation, and genomics tasks
Med-Gemini [19]	2024	Medical language, image, video, EHR	Long-context and search-enhanced medical reasoning	Highlights uncertainty-guided retrieval and broad benchmarks
Generalist biomedical VLM [20]	2024	Biomedical vision–language	Broad biomedical foundation model evaluated across tasks	Supports diverse biomedical image understanding

image–question–answer dataset from biomedical figures [45]. OmniMedVQA and GMAI-MMBench later demonstrated that medical LVLMs still struggle across diverse real medical modalities and perceptual granularities [52], [53].

Medical visual instruction models are attractive because they can answer natural questions, but they also blur evaluation. A model can produce a plausible paragraph that partially answers a question, includes an unsupported finding, and uses medical vocabulary fluently. Automated scoring may mark it correct or near-correct, while a clinician may consider it unsafe. This mismatch is why VQA benchmarks should be complemented with evidence localization, abstention, calibration, and expert harm review.

D. Generalist Biomedical Systems

Med-PaLM M and Med-Gemini represent a shift from single medical VQA models toward generalist biomedical assistants [18], [19]. MultiMedBench combines multiple task types across language, imaging, and genomics. Med-Gemini adds long-context and search-augmented capabilities, reflecting the reality that medical reasoning often depends on lengthy records and external evidence. A generalist biomedical VLM in *Nature Medicine* further shows that broad biomedical image-text pretraining can support diverse tasks [20].

The generalist direction is plausible but not automatically superior. A generalist model may transfer concepts across specialties, but a smaller specialist can be easier to validate, calibrate, and regulate. A hospital may not need a single monolithic model; it may need a federation of specialist encoders, a general language interface, retrieval, structured outputs, and local policy controls. The architectural question is therefore not “generalist or specialist” but how to combine general reasoning with verified specialty evidence.

VI. COMPARATIVE ANALYSIS OF DESIGN TRADE-OFFS

A. Open Models Versus Closed Clinical-Grade Systems

Healthcare MLLM research is split between open academic models and closed industrial systems. Open models such as LLaVA-Med, Med-Flamingo, PMC-VQA-style models, and radiology foundation models allow inspection, replication, local adaptation, and community benchmarking [16], [17], [45], [54]. This openness is scientifically important because clinical claims should be reproducible. It also allows hospitals to test models on local retrospective data without sending patient information to an external service. The weakness is that open models may lag frontier proprietary systems in scale, instruction-following quality, safety tuning, and multimodal breadth.

Closed systems such as GPT-4V, Gemini, GPT-4o, Med-PaLM M, and Med-Gemini often have stronger base capabilities and extensive internal evaluation [8]–[10], [18], [19]. Yet closed weights, undisclosed training data, changing model versions, and limited auditability complicate medical validation. A hospital cannot fully assess benchmark leakage or reproduce a failure if the vendor silently updates the model. For clinical deployment, the question is not whether open or closed is categorically better; it is whether the system provides enough documentation, local validation access, version control, data protections, and incident-response mechanisms for the intended risk class.

B. Generalist Versus Specialist Models

Specialist models can be optimized for a narrow task such as CXR abnormality detection, retinal referral, dermatology triage, or pathology slide classification. Their smaller scope makes validation easier and can yield better calibration on the target distribution. However, specialist tools often fragment

the clinician experience and fail when the task requires cross-modal context. A CXR classifier does not know whether the patient is postoperative; a dermatology image model may not know medication history; a report generator may not know prior imaging.

Generalist biomedical MLLMs aim to unify modalities and tasks. They are attractive for clinical environments where questions are open-ended and evidence is distributed. The risk is that broad capability can conceal uneven performance. A model may perform well on medical multiple-choice questions but poorly on ultrasound images; it may summarize notes well but localize pathology weakly. Generalist systems should therefore publish capability maps rather than a single global score. A clinically credible capability map reports strengths, weak modalities, unsupported tasks, required inputs, and known failure modes.

C. Promptability Versus Reliability

Promptability is one reason MLLMs are compelling. A clinician can ask a new question without retraining a classifier. This flexibility is valuable in education, report QA, and exploratory decision support. In safety-critical contexts, however, promptability can become instability. Small wording changes may alter answer style, uncertainty, or retrieved evidence. A model may comply with an unsafe prompt if role constraints are weak. Clinical systems should use controlled prompt templates for validated tasks, log prompt versions, and separate free-form exploratory use from regulated clinical use.

One practical compromise is to constrain high-risk functions while preserving conversational interaction around low-risk explanations. For example, a CXR system may use a locked report-QA prompt for discrepancy detection, while allowing clinicians to ask educational follow-up questions clearly marked as non-diagnostic. Prompt engineering alone is not a safety strategy; it must be paired with validation, access control, and output constraints.

D. Scale, Context Length, and Evidence Selection

Model scale and context length improve the ability to process long records, but they do not guarantee correct evidence use. Long-context MLLMs may attend to irrelevant historical notes, outdated medication lists, or copied-forward diagnoses. Retrieval can improve currency and traceability, yet retrieval errors create new failure modes. The design problem is evidence selection under clinical uncertainty. A useful system should separate patient-specific evidence, prior imaging evidence, institutional policy, and public literature. It should also show which sources materially influenced the answer.

Context should be treated as a curated clinical object, not a dumping ground. For many tasks, a compact structured summary of recent vitals, medications, labs, and prior reports may be safer than a raw 200-page chart. Future systems should learn to ask for missing context rather than fill gaps. “No prior study was provided” is often a better answer than an unsupported comparison.

E. Cost, Latency, and Workflow Fit

MLLMs compete with clinical time. A model that takes too long, produces verbose answers, or requires complex interaction will not be used in high-volume workflows. Radiology, emergency care, and intensive care need low-latency, concise, interruptible interfaces. Pathology and tumor boards may tolerate longer analysis if evidence is well organized.

Cost also shapes equity: a system that depends on expensive proprietary inference may widen gaps between health systems. Deployment should therefore match model architecture to task tempo. Fast specialist encoders can screen or precompute findings. A larger MLLM can be reserved for synthesis, explanation, or complex cases. Caching, local inference, and retrieval indexes can reduce latency. The human interface should make model uncertainty visible without forcing clinicians to read a long generated essay.

VII. Dataset Analysis

A. Radiology Datasets

Radiology dominates medical MLLM research because images and reports are naturally paired. MIMIC-CXR contains 227,835 imaging studies and 377,110 images with de-identified free-text reports [55]. CheXpert contains 224,316 chest radiographs labeled for 14 observations with uncertainty labels extracted from reports [41]. ChestX-ray8 and PadChest broaden CXR classification resources, while MIMIC-CXR-JPG improves accessibility for machine learning pipelines [55]–[57]. RadGraph provides entity-relation annotations for radiology reports, enabling factuality evaluation beyond surface text [44]. Chest ImaGenome links scene graphs and chest radiographs, supporting structured visual reasoning [58]. These resources are foundational but institutionally skewed; a model trained on them can fail under different scanners, protocols, demographics, or reporting styles.

B. Medical VQA Datasets

VQA-RAD is a manually constructed radiology VQA dataset with clinician-generated questions [59]. PathVQA extends VQA to pathology images [60]. SLAKE adds bilingual questions, semantic labels, and a knowledge graph [61]. PMC-VQA scales VQA using biomedical figures and generated question–answer pairs [45]. OmniMedVQA collects images from 73 datasets spanning 12 modalities and more than 20 anatomical regions, exposing the limited medical generalization of current LVLMs [52]. MIMIC-Ext-MIMIC-CXR-VQA and related PhysioNet resources construct larger CXR VQA datasets from MIMIC-derived labels and templates [62].

The core trade-off is quality versus scale. Expert-generated datasets are clinically meaningful but small. Automatically generated datasets are large but can inherit labeler errors, templated language, and teacher-model artifacts. A robust training strategy should combine both: large weakly supervised corpora for coverage, expert-reviewed sets for validation, and adversarially designed cases for failure-mode analysis.

TABLE IV
COMPARATIVE DESIGN TRADE-OFFS FOR HEALTHCARE MLLMS

Trade-off	Advantage of first pole	Advantage of second pole	Practical recommendation
Open vs. closed	Reproducibility, local testing, inspectable adaptation	Stronger frontier capabilities, extensive vendor tuning	Require model cards, version pinning, local validation, and clear data-use terms regardless of model type
Generalist vs. specialist	Cross-modal reasoning and unified interaction	Easier validation, calibration, and specialty precision	Combine generalist orchestration with validated specialist encoders where risk is high
Promptable vs. constrained	Flexible clinician questions and education	Stable behavior for regulated workflows	Use controlled prompts for validated tasks and separate exploratory use from clinical decision support
Parametric vs. retrieval-grounded	Fast inference and no external dependency	Current, traceable, institution-specific evidence	Use retrieval for guideline- or patient-specific claims and display provenance
Large model vs. small adapter	Broad reasoning, long context, flexible generation	Lower cost, local deployment, easier updates	Match model size to task complexity and monitor cost, latency, and equity
Free-form output vs. structured output	Natural communication and explanation	Interoperability, auditing, downstream automation	Generate both: concise prose plus structured findings, uncertainty, and source links

TABLE V
DATASETS FREQUENTLY USED IN HEALTHCARE MLLM RESEARCH

Dataset	Year	Modalities	Scale/coverage	Typical use	Key limitation
MIMIC-CXR [55]	2019	CXR + reports	227,835 studies; 377,110 images	Image–report pretraining, report generation, retrieval	Single health system; report style and emergency-department distribution
CheXpert [41]	2019	CXR + labels	224,316 radiographs; 14 observations	Classification and uncertainty-label evaluation	Labels extracted from reports; limited findings
ChestX-ray8 [56]	2017	CXR + labels	108,948 images; 8 disease labels	CXR classification	NLP-derived labels and frontal CXR focus
PadChest [57]	2020	CXR + reports/labels	160k+ images; Spanish reports	Multi-label CXR classification and domain shift	Reporting language and institutional differences
VQA-RAD [59]	2018	Radiology images + QA	315 images; 3,515 questions	Clinician-style medical VQA	Small scale; limited modalities
PathVQA [60]	2020	Pathology images + QA	30k+ questions	Pathology VQA	Educational image bias; limited clinical workflow context
SLAKE [61]	2021	Medical images + bilingual QA + labels	642 images; 14k QA pairs	Knowledge-enhanced MedVQA	Moderate scale; curated setting
MedICaT [40]	2020	Biomedical figures + captions/references	217k figures	Scientific image–text pretraining	Figures differ from clinical images
PMC-VQA [45]	2023	Biomedical figures + QA	149k images; 227k QA pairs	Visual instruction tuning for MedVQA	Generated QA quality varies; publication-image bias
OmniMedVQA [52]	2024	Medical images + QA	73 datasets; 12 modalities	Broad LVLM evaluation	Benchmark design still cannot replace prospective clinical validation
MIMIC-Ext-CXR-VQA [62]	2024	CXR + VQA + EHR-derived resources	Approx. 377k entries	CXR VQA and EHR-image QA	Template and labeler dependencies
RadGraph [44]	2021	Radiology reports + entity relations	Expert-labeled dev/test; large inferred set	Factual report evaluation and graph extraction	Text-graph focus; image grounding indirect

C. Biomedical Figure and Literature Datasets

MedICaT contains medical images, captions, and textual references from open biomedical literature [40]. PubMed and PubMed Central supply biomedical language and figure-caption data for models such as BioBERT, PubMedBERT, BioGPT, BiomedCLIP, and LLaVA-Med [16], [34], [35], [37], [51], [63]. Scientific figures are valuable because they cover many modalities and specialties, but they differ from clinical images: they may be composites, annotated panels, plots, or curated exemplars. A system trained heavily on literature figures may perform well on biomedical educational images but poorly on routine clinical DICOM studies.

VIII. BENCHMARK AND EVALUATION ANALYSIS

A. Why Conventional Metrics Are Insufficient

Medical MLLMs are often evaluated with accuracy, BLEU, ROUGE, CIDEr, BERTScore, AUC, exact match, or expert preference [64]–[67]. These metrics are useful but incomplete. BLEU and ROUGE reward lexical overlap, yet two clinically equivalent reports can use different wording, and one omitted word can change a diagnosis. AUC summarizes discrimination but not calibration, subgroup fairness, or clinical threshold choice. VQA exact match is simple but brittle for free-form answers. LLM-as-judge evaluation is scalable but can inherit evaluator bias and may miss subtle medical errors.

Clinical evaluation should be multi-layered. At minimum, a healthcare MLLM should be assessed for (i) task accuracy; (ii) factual consistency with input images and patient data; (iii) uncertainty calibration; (iv) robustness to acquisition,

demographic, and linguistic shift; (v) evidence grounding; (vi) safety under adversarial or ambiguous prompts; (vii) workflow impact; and (viii) user trust and automation-bias effects. A model that improves average VQA accuracy while increasing unsupported diagnostic assertions is not clinically better.

B. Report Generation Evaluation

Radiology report generation exposes the metric problem sharply. Early systems optimized language overlap, but clinical utility depends on whether findings are complete and consistent. CheXbert and CheXpert-label F1 compare disease labels extracted from generated and reference reports [41], [43]. RadGraph evaluates entities and relations, capturing whether an observation is present, absent, uncertain, and linked to anatomy [44]. Factual reward methods encourage entity completeness and natural-language-inference consistency [68]. Human radiologist review remains essential because automated labelers cannot judge every clinically important nuance.

C. Calibration and Uncertainty

Calibration matters because a clinician may act differently on a model's answer if it is uncertain. For predicted confidence $\hat{p}$ and correctness y, expected calibration error (ECE) partitions predictions into bins B_m:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|. \tag{11}$$

Generative MLLMs complicate calibration because token probabilities do not directly equal clinical confidence. Alternatives include verbalized uncertainty, self-consistency, retrieval agreement, conformal prediction, abstention thresholds, and expert-reviewed uncertainty labels. However, verbalized uncertainty can be poorly calibrated; a model may say "likely" for both common and rare errors. Calibration should be evaluated empirically by task, site, subgroup, and prompt style.

D. Benchmark Contamination and Leakage

Biomedical benchmarks are vulnerable to leakage because public datasets, reports, and benchmark answers may appear in web-scale pretraining corpora. Closed models make contamination difficult to audit. Even when direct leakage is absent, templated question generation can make train and test distributions too similar. Medical benchmarks should therefore publish contamination analyses, maintain hidden test sets when appropriate, and include expert-designed cases that test reasoning rather than memorized associations.

IX. Applications and Case Studies

A. Medical Visual Question Answering

Medical VQA tests whether a model can connect perception, clinical knowledge, and language. In radiology, a question may ask about modality, anatomy, abnormality, location, severity, comparison, or next-step implication. In pathology, the model may need to identify tissue type or histologic pattern. In dermatology or ophthalmology, color, texture, and lesion boundaries matter. MLLMs are attractive because they can answer open-ended questions without a separate classifier for each label.

However, VQA should not be confused with diagnosis. Many datasets ask short, image-contained questions, while real clinical questions require history, priors, labs, and treatment context. A safe VQA assistant should answer only the visual question when the image is sufficient, state missing context when it is not, and distinguish educational explanations from clinical recommendations. Expert evaluation should score not only correctness but also whether the model overreaches.

B. Radiology Report Drafting and Quality Control

Report generation is the most mature MLLM healthcare application because CXR datasets naturally pair images and reports. A practical near-term workflow is not autonomous reporting but report assistance. The model may generate an initial findings draft, summarize priors, flag discrepancies between dictated text and image evidence, or convert free text into structured entities. The radiologist remains responsible for review and finalization.

Quality-control use cases may be safer than full generation. For example, an MLLM can compare a draft report with image-derived findings and ask, "The draft mentions no support devices, but a right internal jugular central line may be visible. Please review." Such a system still requires validation, but it acts as a second reader rather than a replacement author. It also produces auditable prompts for human correction.

C. Pathology, Ophthalmology, and Dermatology

Pathology MLLMs face extreme image scale and weak labels. Whole-slide diagnosis depends on tissue architecture, cellular morphology, stain quality, and clinical context. MLLMs can support slide triage, educational question answering, report summarization, and molecular-test integration, but they require hierarchical encoders and strong grounding. Ophthalmology and dermatology are attractive because imaging is standardized in some tasks, yet population diversity, device variation, and acquisition quality strongly affect performance [69]–[71]. Patient-facing dermatology assistants are especially risky because lighting, skin tone, camera artifacts, and differential diagnosis can alter safety.

D. EHR Summarization and Longitudinal Reasoning

EHR summarization is primarily language and structured-data oriented, but it becomes multimodal when reports, images, waveforms, and labs are integrated. Long-context MLLMs can summarize admissions, produce discharge drafts, and retrieve relevant prior findings. The central challenge is temporal reasoning: a model must know whether a lab abnormality is current, resolved, chronic, or medication-induced. It must also preserve source provenance. A summary that merges old and new conditions without dates can mislead clinicians.

TABLE VI
EVALUATION LAYERS FOR HEALTHCARE MLLMS

Layer	Example metrics	What it captures	What it misses
Language overlap	BLEU, ROUGE, CIDEr	Surface similarity for generated reports or summaries	Factual correctness, clinical severity, synonymy, omissions
Semantic similarity	BERTScore, embedding similarity	Paraphrase-aware similarity	Negation, uncertainty, anatomy-specific relations
Clinical labels	CheXpert/CheXbert F1, AUC	Disease-label agreement and classification performance	Fine-grained findings, devices, measurements, differential diagnosis
Graph/entity metrics	RadGraph F1, relation consistency	Structured clinical facts and relations	Visual evidence and broader context
VQA accuracy	Exact match, relaxed accuracy, expert score	Short-answer correctness	Explanation quality, harm, uncertainty, partial correctness
Grounding	IoU, pointing game, phrase-region accuracy	Evidence localization	Whether localization caused the answer; text evidence relevance
Calibration	ECE, Brier score, abstention curves	Confidence reliability	Free-form uncertainty wording and user interpretation
Clinical workflow	Time saved, error recovery, prospective outcomes	Real-world utility and human-AI interaction	Hard to generalize; costly to measure

TABLE VII
CLINICAL APPLICATION MATRIX

Application	MLLM contribution	Required evidence	Deployment caution
Radiology VQA	Answers image-specific clinician questions and explains visual findings	Image, report context, prior studies when comparison is requested	Avoid diagnostic overreach; require uncertainty and image grounding
Report drafting	Generates findings/impression drafts or structured report entities	DICOM image, prior reports, indication, report templates	Human sign-off; factuality and omission checks essential
Report QA	Flags contradictions, missing devices, laterality errors, or unsupported statements	Draft report, image evidence, structured labelers/graphs	Alert fatigue and false positives must be measured
Pathology assistance	Summarizes WSI regions, answers morphology questions, links to molecular context	Slide patches, report, stains, molecular tests	Requires hierarchical grounding and expert review
Ophthalmology/dermatology triage	Describes lesions or retinal findings and recommends urgency categories	High-quality images, demographics, symptoms, history	Bias across skin tone/device/site; patient-facing safety controls
ICU multimodal summarization	Synthesizes notes, waveforms, vitals, labs, medications, imaging	Longitudinal EHR, monitor data, medication timeline	Temporal errors can be dangerous; provenance required
Patient education	Converts findings into accessible language and answers follow-up questions	Final clinician-approved report and vetted educational content	Should not replace clinician consultation or alter treatment plans

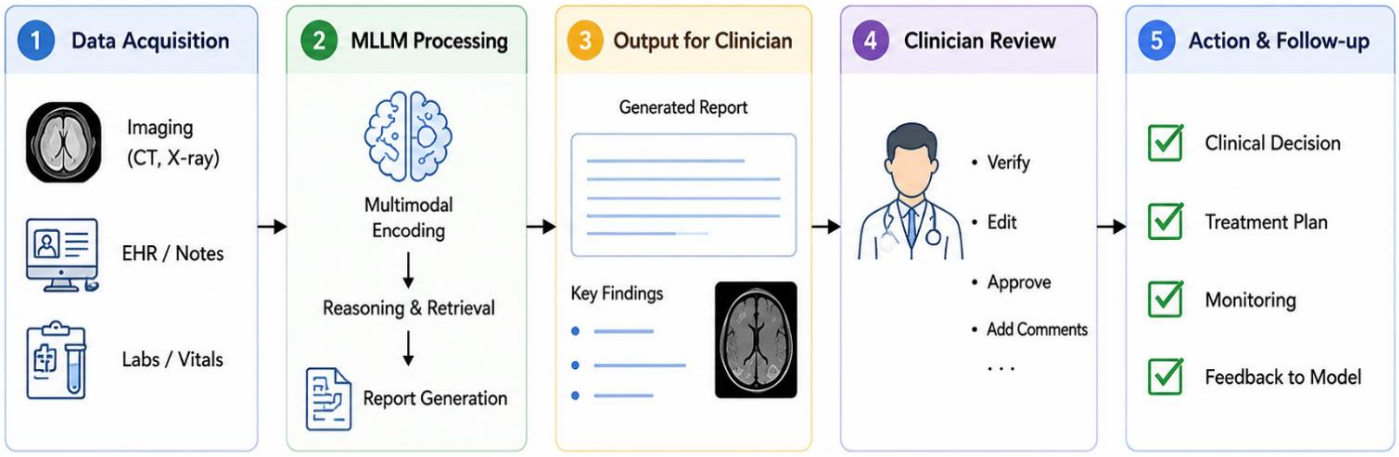

Fig. 4. illustrates an end-to-end clinical workflow where healthcare MLLMs process multimodal patient data, including medical imaging, EHR notes, and laboratory records, to generate clinically interpretable outputs. The workflow emphasizes clinician-in-the-loop review, ensuring human oversight, validation, and evidence-based decision support before final clinical action.

E. Patient Communication and Medical Education

MLLMs can translate radiology impressions into plain language, create patient-specific explanations, simulate teaching cases, and support medical trainees. These applications have lower immediate risk than autonomous diagnosis if they are constrained and reviewed. Even here, outputs must be calibrated. A patient-facing explanation should not minimize serious uncertainty, and an educational answer should distinguish canonical textbook reasoning from patient-specific advice.

F. Deployment Case Study: CXR Reporting Copilot

Consider a CXR reporting copilot deployed inside a picture archiving and communication system. The model receives the current frontal and lateral images, the indication, the prior report, and optionally prior images. It produces a structured draft with findings, impression, uncertainty flags, and evidence pointers. A safety layer checks for unsupported claims using labelers, RadGraph-style entity consistency, and retrieval of prior reports. The radiologist edits the draft, rejects unsafe suggestions, and signs the final report. Model outputs and human edits are logged for monitoring.

This workflow illustrates why clinical deployment is a system problem. The visual encoder may miss a small pneumothorax; the LLM may generate a normal impression inconsistent with a finding; the retrieval layer may select the wrong prior; the interface may encourage over-trust; and the monitoring system may miss subgroup drift. A validated copilot must evaluate every component plus the human-AI team.

G. Deployment Case Study: Multimodal Triage Assistant

In emergency triage, an MLLM could summarize chief complaint, vitals, labs, ECG, CXR, and prior history to suggest urgency or missing data. The model should not independently assign final triage without nurse or physician review. It should instead highlight evidence: “oxygen saturation decreased from 96% to 90%, CXR report mentions new bilateral opacities, and troponin is pending.” The output should be concise, source-linked, and uncertainty-aware. Evaluation should measure whether clinicians detect more critical cases, whether delays decrease, and whether false alarms create workload.

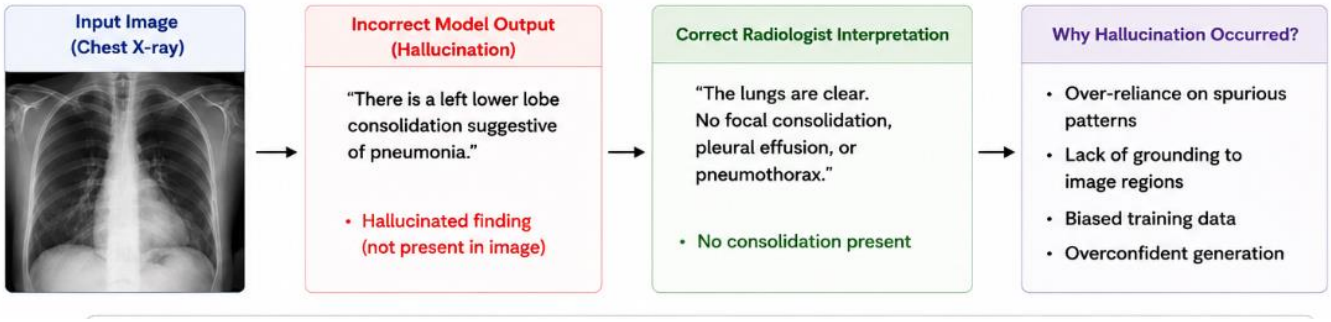

Fig. 5. Illustration of hallucination in a healthcare MLLM where the model generates an incorrect radiological finding that is not present in the input chest X-ray. The figure highlights the importance of visual grounding, clinician verification, and uncertainty-aware reasoning in medical AI systems.

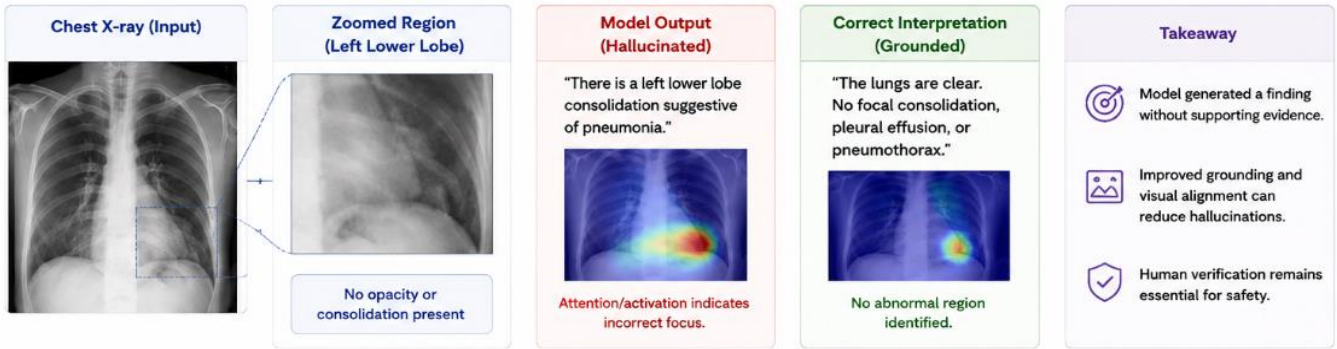

Fig. 6. This figure demonstrates how a medical MLLM may incorrectly focus on irrelevant image regions and produce unsupported clinical interpretations. The comparison between hallucinated and grounded outputs emphasizes the need for robust evidence alignment and human oversight in safety-critical healthcare applications.

X. Challenges and Limitations

A. Hallucination and Unsupported Reasoning

Hallucination is the central barrier to clinical trust. In healthcare MLLMs, hallucination can mean inventing a finding, fabricating patient history, citing a nonexistent guideline, misreading laterality, or producing an unsupported treatment suggestion. Object hallucination has been studied in general captioning [72]; medical hallucination is more consequential because it can alter decisions. Grounded generation, retrieval, structured outputs, constrained decoding, and abstention can reduce hallucination but cannot eliminate it. Clinical systems must assume residual errors and design review workflows accordingly.

B. Bias and Distribution Shift

Medical data encode social and institutional bias. Historical undertreatment, access disparities, scanner distribution, report style, and demographic underrepresentation can all influence model behavior [73], [74]. Distribution shift occurs when a model trained at one site is deployed at another with different patient populations, imaging protocols, device vendors, or language patterns. MLLMs add prompt and retrieval shift: the same image can receive different answers under different instructions or retrieved context. Robust evaluation must therefore include external sites, subgroup analysis, acquisition variation, and prompt sensitivity.

C. Explainability and Grounding

Clinicians often ask why a model produced an answer. Attention maps, saliency, region pointers, retrieved cases, and natural-language rationales can help, but each can mislead. A heatmap may highlight a correlated artifact rather than a causal finding. A rationale may be generated after the answer and not reflect internal computation. Grounding should therefore be treated as an empirical claim: does the highlighted region or cited sentence actually support the answer, and does removing it change the output? Strong explanation combines visual grounding, textual provenance, uncertainty, and human factors testing.

D. Privacy, Security, and Memorization

Healthcare MLLMs can leak protected health information through training memorization, prompt logs, retrieval caches, or generated summaries. Multimodal data increase the attack surface: images may contain burned-in identifiers, waveforms may be linkable, and rare cases may be identifiable even after de-identification. Security risks include prompt injection, retrieval poisoning, adversarial images, model inversion, and unauthorized tool calls. Privacy-preserving training helps but must be paired with access control, logging, redaction, secure deployment, and incident response.

XI. Ethical, Legal, and Regulatory Considerations

A. Clinical Accountability

MLLMs do not remove professional accountability. A safe deployment must define who reviews outputs, who can override them, when model use is documented, and how errors are reported. Responsibility is distributed across developers, health systems, clinicians, procurement teams, and regulators. Ambiguity is dangerous: if a model's output affects care, the workflow must specify whether it is advisory, diagnostic, administrative, or patient-facing.

B. Regulatory Pathways

In the United States, AI/ML-enabled Software as a Medical Device (SaMD) is addressed through FDA guidance, including the AI/ML SaMD Action Plan, Good Machine Learning Practice, transparency principles, and change-control planning [22], [75]–[77]. The European AI Act classifies many medical AI systems as high-risk, requiring risk management, high-quality data, transparency, human oversight, and post-market monitoring [23]. The WHO guidance on large multimodal models emphasizes governance, stakeholder involvement, transparency, safety, cybersecurity, and equity [21]. These frameworks are evolving, but their shared message is stable: clinical AI must be governed across the total product life cycle, not only at model release.

C. Ethics of Data and Consent

Medical MLLMs are trained on data produced in care, research, or publication contexts. Patients may not expect their scans or notes to train commercial foundation models. Public biomedical figures may be open access, but patient images inside publications still carry consent and dignity considerations. Survey authors and model developers should document data provenance, licenses, consent assumptions, and exclusion criteria. Dataset datasheets and model cards are not paperwork; they are tools for accountability [78], [79].

D. Human-Centered Deployment

Human-centered deployment means that interfaces should support clinical reasoning rather than replace it. The best interface may not be a chat window. It may be a side panel that lists evidence, displays uncertainty, highlights discrepancies, and lets clinicians accept or reject structured suggestions. Alert design should minimize fatigue. Patient-facing outputs should be constrained to approved use cases and written in accessible language. Training should teach clinicians both capabilities and failure modes.

XII. Future Directions

A. Grounded Generalist–Specialist Collaboration

Future systems will likely combine generalist language reasoning with specialist encoders and tools. A generalist model can coordinate tasks, interpret user intent, and communicate results, while specialist models provide calibrated predictions

TABLE VIII
RISK ANALYSIS FOR HEALTHCARE MLLMS

Risk	Failure example	Consequence	Mitigation direction
Visual hallucination	Reports a pneumothorax absent from CXR	Unnecessary treatment or alarm	Grounded decoding, expert review, abstention, report-image consistency checks
Omission	Misses a subtle line malposition or small effusion	Delayed diagnosis or intervention	High-sensitivity QA layer, second-reader workflow, targeted rare-finding tests
Temporal confusion	Treats old pulmonary edema as current	Wrong assessment of disease trajectory	Time-stamped context, prior/current separation, temporal encoders
Bias	Lower accuracy for underrepresented demographic or device group	Unequal care quality	Subgroup evaluation, external validation, fairness monitoring
Retrieval error	Cites outdated or irrelevant guideline	Misleading recommendation	Source ranking, version control, provenance display, clinician review
Privacy leakage	Reveals identifiable prior case details	Legal and ethical harm	De-identification, access control, DP/federation, prompt-log governance
Automation bias	Clinician accepts fluent wrong answer	Reduced error detection	Interface design, uncertainty display, training, forced evidence review
Regulatory drift	Model updates after approval without monitoring	Uncontrolled performance change	Change-control plans, post-market surveillance, locked validated versions

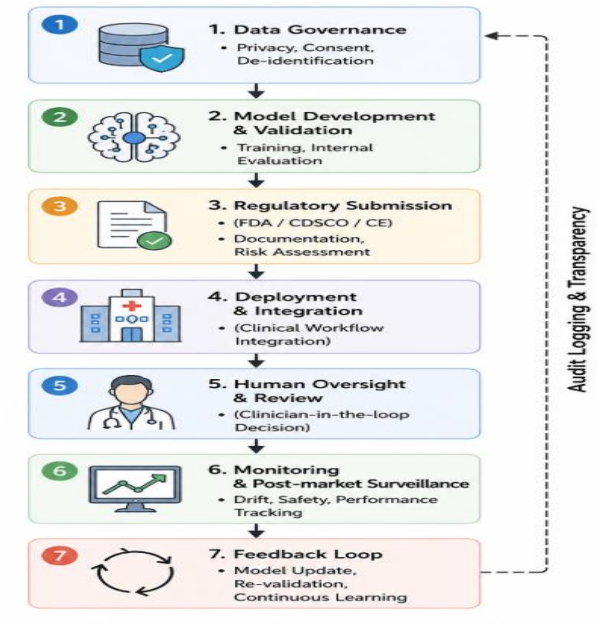

Fig. 7. Author-created governance workflow for clinical MLLM deployment. The key principle is life-cycle control: validation, monitoring, incident review, and model updates must be planned before deployment.

for CXR, pathology, ECG, dermatology, or ophthalmology. Retrieval systems can add guidelines and patient-specific priors. This design is more auditable than a monolithic assistant and aligns with clinical practice, where generalists consult specialists and evidence sources.

B. Multimodal Temporal Reasoning

Temporal reasoning remains underdeveloped. Healthcare decisions depend on change: interval growth, lab trends, medication response, postoperative evolution, and disease progression. Future MLLMs need time-aware encoders that distinguish current findings from history, support comparison across studies, and represent uncertainty over time. Benchmarks should include temporal tasks such as “has edema improved compared with yesterday?” or “is this troponin pattern consistent with acute injury?”

C. Clinically Meaningful Benchmarks

Benchmarks should move beyond static leaderboards. Needed evaluations include hidden expert-reviewed test sets, subgroup analysis, out-of-distribution sites, rare but high-harm findings, adversarial prompt suites, retrieval-conflict cases, and prospective workflow studies. Evaluation should be reported with confidence intervals and stratified performance, not only a mean score. For generative outputs, factuality, calibration, and evidence grounding should be first-class metrics.

D. Privacy-Preserving Collaborative Training

No single institution has sufficient diversity to train and validate robust healthcare MLLMs. Federated and privacy-preserving approaches are therefore important, but they must be engineered for multimodal scale. Research is needed on federated instruction tuning, secure multimodal retrieval, private adapter sharing, differential privacy for generative models, and evaluation of privacy-utility trade-offs. Hospitals should be able to participate in model improvement without relinquishing governance over patient data.

E. Sustainable and Resource-Aware Deployment

Large MLLMs consume substantial compute and energy. Healthcare deployment should match model size to task risk and complexity. A small specialist model may be preferable for high-volume screening; a large MLLM may be reserved for complex summarization or cross-modal reasoning. Compression, caching, retrieval specialization, and edge inference will be necessary for resource-constrained settings. Equity also requires that the strongest tools not be available only to wealthy institutions.

XIII. CONCLUSION

MLLMs mark a major transition in healthcare AI from isolated classifiers toward interactive systems that can see, read, retrieve, summarize, and communicate. The field has progressed rapidly from contrastive image–report representation learning to instruction-tuned biomedical assistants and

generalist systems spanning multiple tasks and modalities. Yet clinical readiness remains limited by hallucination, weak grounding, dataset shift, bias, privacy risk, uncertain calibration, and evaluation methods that often reward benchmark performance rather than patient safety.

The path forward is neither blind enthusiasm nor rejection. Healthcare MLLMs can be valuable if they are designed as governed clinical systems: grounded in patient evidence, evaluated across sites and subgroups, constrained by role and use case, integrated into human workflows, and monitored across the product life cycle. The strongest future systems will likely combine generalist reasoning, specialist models, retrieval, structured outputs, and transparent governance. In medicine, intelligence is not enough; evidence, accountability, and humility must be engineered into the system.

ACKNOWLEDGMENT

The authors acknowledge the Department of Computer Science Engineering, Bennett University, for supporting the original undergraduate research draft. The authors also acknowledge the open research community whose papers, datasets, model cards, and public benchmark reports made this survey possible.

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